Counting.

Discrete combinatorial systems.

Sum Rule:

Two tasks:

First task can be done in n ways

Second task can be done in m ways

tasks cannot be done simultaneously

or are mutually exclusive.

then performing either task can be accomplished in $m+n$ ways.

E.g. 1. Student Antropology — sociology (40 text books), Antropology (50 text books)

$40 + 50 = 90$ textbooks to learn about one or the other of these two subjects

E.g. 1.2.

Student of CS.

↳ first programming language.

APL — 5
BASIC — 5
FORTRAN — 5
PASCAL — 5

$5+5+5+5 = 20$ books available for someone interested in learning intro to programming.

Product rule of counting

* The task must be complete in more than one stage(s).

m possible outcomes from the first stage
n possible outcomes from each of them m
then the total outcomes is $m \times n$

$\boxed{\text{"Prinple of choice"}}$

Eg. 1.3.

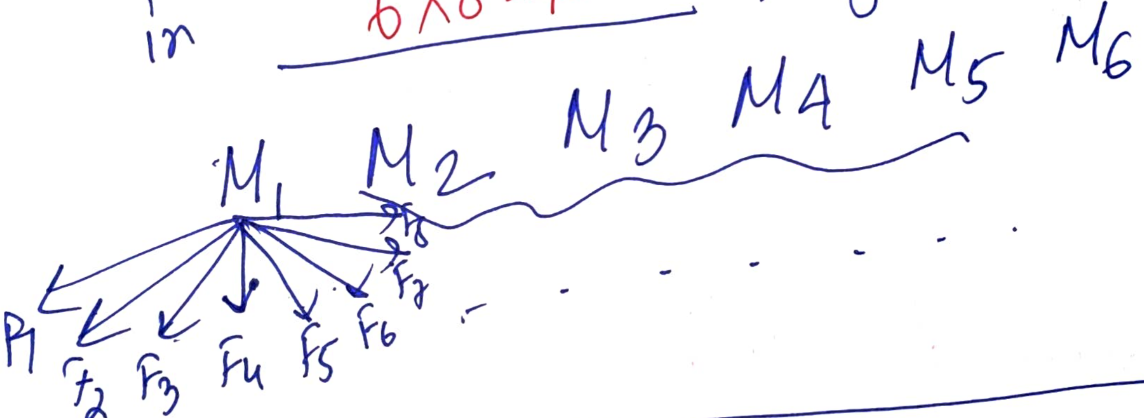
Drama club.

Lead roles

males → (6 candidates)

female → (8 ~~female~~ candidates)

The director can cast the leading role in $\underline{6 \times 8 = 48}$ ways.



48 ways.

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information → computers → memory.

memory → arranged in units → each unit → memory cell.

memory cell → name → address of the cell.

address of the cell → composed of bits (binary digits) → 0, 1

(*) 8 bits are used to represent a single address.

8 bits $\longrightarrow$ 1 byte.

$\downarrow$ How many addresses can you represent?

$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^8 = 256$.

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

PDP-11 $\longrightarrow$ 2byte addressing scheme.

How many addresses? 2^{16}

Modern day 64 bit $\longrightarrow$ no. of address location 2^{64}.

Count $\longrightarrow$ linear arrangements of distinct objects $\longrightarrow$ permutation.

Eg. 1.4

Class of 10 students.

5 of them in a row to take a picture. How many linear arrangements are possible?

1 2 3 10

A B C . . . . J.

→ Arrangement : $\boxed{\text{BCEFI}}$

ABCFG

.

$\boxed{\text{CEFIB}}$

same set of students

distinct pictures.

order in which students sit make the photographs distinct.

In counting when order is necessary → permutation creeps in.

$$\left[\begin{matrix}10\\ \text{1st position}\end{matrix}\right] \times \left[\begin{matrix}9\\ \text{2nd position}\end{matrix}\right] \times \left[\begin{matrix}8\\ \text{3rd position}\end{matrix}\right] \times \left[\begin{matrix}7\\ \text{4th position}\end{matrix}\right] \times \left[\begin{matrix}6\\ \text{5th position}\end{matrix}\right]$$

$$= \frac{10!}{5!} = 10 \times 9 \times 8 \times 7 \times 6.$$

<u>Permutation:</u> If there are n distinct objects $a_1 \ldots a_n$ and r is an integer $1 \le r \le n$ then by the rule of product, the no. of permutations of size r of the n objects is $\boxed{n \times (n-1) \times (n-2) \times \cdots (n-r+1) = \frac{n!}{(n-r)!} = P(n,r)}$.

Eg. 1.5. Count all the arrangements of all the six letters in PEPPER.

$P_1 E_1 P_2 P_3 E_2 R$.

↳ no. of arrangements $= 6!$

$$\left.\begin{array}{l} P_1 E P_2 P_3 E R \\ P_1 E P_3 P_2 E R \\ P_2 E P_1 P_3 E R \\ P_2 E P_3 P_1 E R \\ P_3 E P_1 P_2 E R \end{array}\right\} \rightarrow \text{all same.}$$

$3!$ – "sames"

$$\left.\begin{array}{l} P_1 E_1 P_2 P_3 E_2 R \\ P_1 E_2 P_2 P_3 E_1 R \end{array}\right\} \rightarrow \text{same}. \quad 2! - \text{"sames"}$$

$2! \times 3! \times$ (No. of arrangements of the letters in PEPPER)

= Number of permutation of the symbols $P_1, E_1, P_2, P_3, E_2, R = 6!$

No. of actual arrangements $= \boxed{\dfrac{6!}{2!\,3!}}$

If there n objects with n_1 of first type, n_2 of second type. ... , n_r of r-th type

s.t. $n_1 + n_2 + \dots + n_r = \underline{n}$

then there are $\dfrac{n!}{n_1!\,n_2!\cdots n_r!}$ linear arrangements of the given n objects.

Eg. 1.5.

Find the no. of linear arrangements in the word $\boxed{\text{MASSASAUGA}}$.

$$\frac{10!}{4!\,3!\,1!\,1!\,1!}$$

H.W. In how many of these above case does all the four A's appear together?